

Reproductive period and risk of dementia in postmenopausal women.

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Evidence abstract

CONTEXT Exogenous estrogen use may lower risk of dementia in postmenopausal women.

A relationship between long-term exposure to endogenous estrogens and incident dementia has been hypothesized but not studied.

OBJECTIVE To determine whether a longer reproductive period, as an indicator of longer exposure to endogenous estrogens, is associated with lower risk of dementia and Alzheimer disease (AD) in women who have natural menopause.

DESIGN AND SETTING The Rotterdam Study, a population-based prospective cohort study conducted in the Netherlands.

PARTICIPANTS A total of 3601 women aged 55 years or older who did not have dementia at baseline (1990-1993) and had information on age at menarche, age at menopause, and type of menopause.

Participants were reexamined in 1993-1994 and 1997-1999 and were continuously monitored for development of dementia.

MAIN OUTCOME MEASURES Incidence of dementia, based on Diagnostic and Statistical Manual of Mental Disorders, Revised Third Edition criteria, and AD, based on National Institute of Neurological Disorders and Stroke/Alzheimer's Disease and Related Disorders Association criteria, compared by quartiles of reproductive period among women with natural menopause.

RESULTS During 21 046 person-years of follow-up (median follow-up, 6.3 years), 199 women developed dementia, including 159 who developed AD.

After adjusting for age, dementia was not clearly associated with length of reproductive period.

However, after adjusting for multiple covariates, women with natural menopause and more reproductive years had an increased risk of dementia (adjusted rate ratio [RR] for women with >39 reproductive years [highest quartile] compared with <34 reproductive years [lowest quartile], 1.78; 95% confidence interval [CI], 1.12-2.84).

The adjusted RR per year of increase was 1.04 (95% CI, 1.01-1.08).

For risk of AD, the adjusted RRs were 1.51 (95% CI, 0.91-2.50) and 1.03 (95% CI, 1.00-1.07), respectively.

Risk of dementia associated with a longer reproductive period was most pronounced in APOE epsilon4 carriers (adjusted RR for >39 reproductive years compared with <34 reproductive years, 4.20 [95% CI, 1.97-8.92] for dementia and 3.42 [95% CI, 1.51-7.75] for AD), whereas in noncarriers, no clear association with dementia or AD was observed.

CONCLUSION Our findings do not support the hypothesis that a longer reproductive period reduces risk of dementia in women who have natural menopause.

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